MEG Study of Structural Priming in RPVP

The brain basis of syntax remains debated (Friederici, 2011; Matchin and Hickok, 2020; Fedorenko et al., 2020; Matar et al., 2021). One productive approach for probing the neural basis of different types of representations is priming. Here we used MEG and a structural priming paradigm to test whether syntactic structure persists as a neural representation after exposure to a grammatical expression. In behavior, structural priming effects have been well-documented in language production (Mahowald et al., 2016) are less so in comprehension (Tooley & Traxler, 2010), possibly due to the more passive nature of comprehension (van Lieburg et al., 2023).

Here we explored a new potential explanation for the asymmetry: that the serial nature of sentence processing may lessen the synchronicity of the activation of a syntactic tree representation, and consequently its availability as a primed representation in a priming paradigm. This would happen because in serial presentation, more trees are likely to be considered by the parser, lessening the strength of the "right" tree as the prime representation.

To test whether clear structural priming would be observed when the serial unfolding of language is removed, we combined the Rapid Parallel Visual Presentation (RPVP, Snell & Grainger, 2017) paradigm with a structural priming manipulation in English comprehension. Our two syntactic structures were closely matched in their form properties, yet differing in syntax: Det-Adj-Noun sequences such as 'the green color' and Det-Noun-Noun sequences such as 'the color green.' They were entered into a 2 × 2 factorial design manipulating Syntactic Structure (Same/Different between prime and target) and Lexical Overlap (Same/Different). Color lists such as 'tan green black' were used as non-combinatory controls. MEG signals from thirty-nine participants were collected during a behavioral matching task, following the procedure in Fallon & Pylkkanen (2024) and Flower & Pylkkanen (2024).

We observed both behavioral facilitation and increased neural signals as a function of phrasal structure, consistent with prior reports of so-called Structure Superiority Effects using RPVP. However, even though the presence of syntactic structure robustly enhanced neural signals across left lateral language cortex on the prime, these signals did not show reliable priming as a function of shared syntactic structure, only as a function of shared lexical material, in the superior temporal cortex (BA22). Thus, we found no compelling evidence that syntactic structures linger as primable neural representations in short-term memory, independent of the words that express them. This broadly converges with the prior syntactic priming literature, which has rarely reported pure syntactic priming effects in comprehension in the absence of lexical overlap. Nonetheless, our behavioral matching task did reveal a new type of syntactic priming effect: when detecting mismatch between two stimuli, mismatching syntactic structures facilitated No-responses. Therefore, even though our syntactic manipulation was very subtle, involving no surface differences in the stimuli, participants clearly computed the underlying structures automatically, and this affected their behavioral responses.

References

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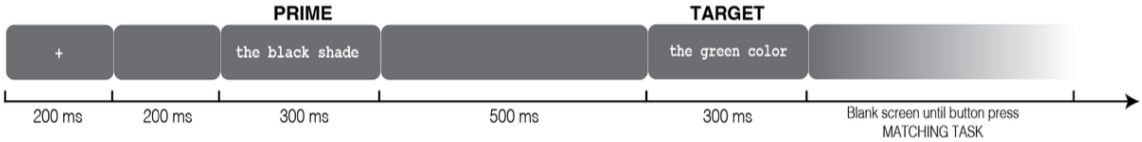
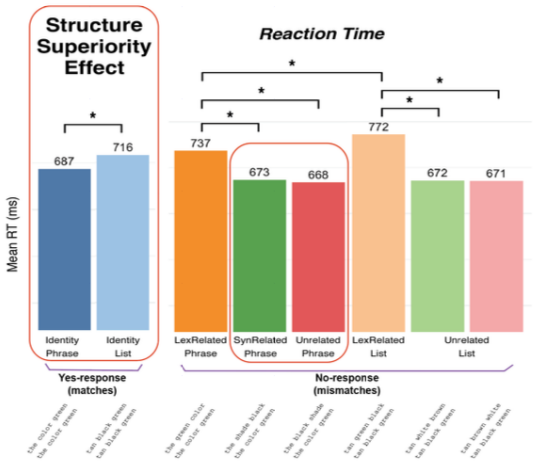
Trial Structure



Design and reaction time results

Composition	Condition	Prime	Target	Match?
Phrase	Identity	the color green	the color green	Yes
	LexRelated (DiffStruc)	the green color	the color green	No
	SynRelated (DiffOrder)	the shade black	the color green	No
	Unrelated	the black shade	the color green	No
List	Identity	tan black green	tan black green	Yes
	LexRelated (DiffOrder)	tan green black	tan black green	No
	Unrelated	tan white brown	tan black green	No
		tan brown white	tan black green	No

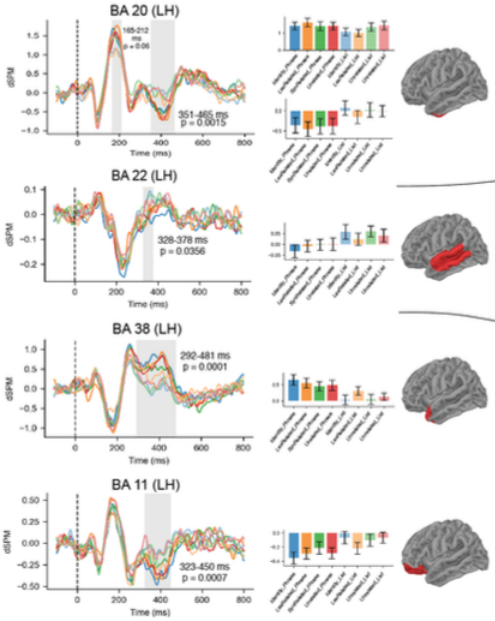


MEG results

PRIME: Structure Superiority Effect across fronto-temporal cortex

Temporal clustering anova with permutations, in language ROIs (BA11 20 21 22 37 38 39 44 45 47, with FDR correction)

Search window: 100-600ms — Prime locked

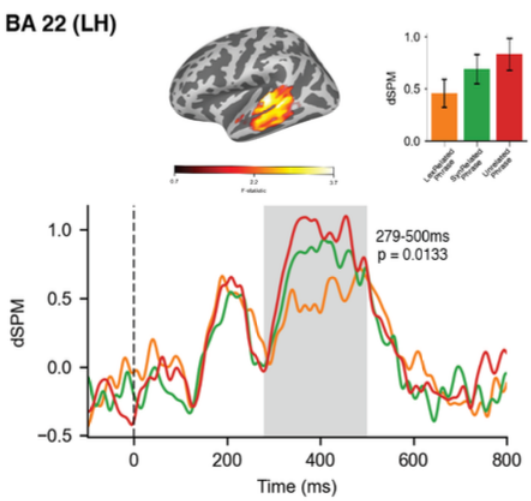


TARGET: Priming in BA22

From the regions showing evidence of Structure Superiority on the Prime, only BA22 showed a priming effect on the Target, with unrelated targets showing the highest amplitude at ~300-500ms, lexically related targets the lowest, and syntactically related targets patterning in the middle.

Spatiotemporal clustering anova with permutations

Search window: 100-600ms — Target locked



In pairwise comparisons, Lexically related and Unrelated targets were significantly different (p = .0001) but Syntactically related targets did not reliably diverge from either (SynRelated vs Unrelated p = .24, LexRelated vs SynRelated p = .11).